Nanyang Technological University
School of Electrical & Electronic Engineering
IE2106 Engineering Mathematics I

Tutorial 9 Numerical Methods

1. Consider the initial value problem

$$y' = \frac{1}{2}\left(\frac{y}{x} - \frac{x}{y}\right), \qquad y(2) = 2.$$

Use the following methods with step size $h = 0.2$ to find $y(2.6)$.

(a) Euler method

(b) Improved Euler method

(c) Runge-Kutta method

(Exact solution is $y = \sqrt{4x - x^2}$)

2. Consider the initial value problem

$$y'' + 2(y')^2 + 10y = 0$$

$$y(0) = 0, \qquad y'(0) = 1.$$

Use Euler method with step size $h = 0.1$ to approximate $y(0.2)$ and $y'(0.2)$.

3. Consider the initial value problem

$$y''(x) - xy'(x) + y(x) = x^2$$

$$y(2) = 3, \qquad y'(2) = 0$$

where $y''(x) = \frac{d^2y}{dx^2}$ and $y'(x) = \frac{dy}{dx}$. Use improved Euler method (Heun's method) with step size $h = 0.1$ to find approximate values of $y(2.3)$ and $y'(2.3)$.

4. Apply Runge-Kutta method with step size $h = 0.04$ to the initial value problem (here $y'' = \frac{d^2y}{dt^2}$)

$$y'' = 5 + 2ty$$

$$y(0) = 2, \qquad y'(0) = 0.$$

Find approximate values of $y(0.12)$ and $y'(0.12)$.

Answers

1. (a) 1.942290 (b) 1.909421 (c) 1.907879

2. 0.18, 0.572

3. 3.0731960 , 0.646850811

4. 2.0371, 0.6290